

9/02/26 (Monday)

* Hill Climbing :

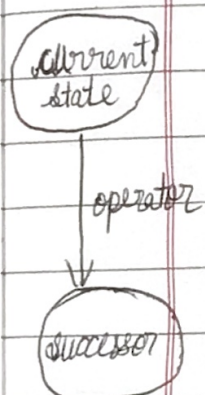
- ⇒ generator assign the direction to move in search space
- ⇒ terminate when you reach the goal state.
- ⇒ maximum / minimum ~~value~~ problem.

* simple hill climbing algorithm :-

- ① Evaluate the initial state. If it is the goal state then quit otherwise continue with initial state as current state.
- ② Loop until solution is found or no new operators are left to apply.
 - (a) select an operator that has not yet applied and apply to produce a new state.
 - (b) If it is not a goal state but it is better than current state then Make it current state.
 - (c) If it is not better than current state then continue in the loop.

* Steepest descent Hill Climbing (Gradient search):

- ① evaluate the initial state. If it is the goal state then quit otherwise continue with the initial state as current state.
- ② Loop until solution is found or complete iteration produces no change to current state.
 - (a) let successor be a state such that any possible successor of the current state will be better than the successor.
 - (b) for each operator that applies to current state:
 - apply operator and generate a new state
 - examine the new state & if it is GOAL state then quit.
 - if it is better than successor then update successor otherwise continue
 - (c) if successor is better than current state ^{then} make current state successor.



NOTE : It may be possible that both the algorithms may not give the required solution. Because (Reason):

(i) if state fails at Local Maximum: if its current state is better than the neighbour states but the better state is far away from C.S.,



(ii) Plateau: if all the neighbours ^{state almost} have same values as C.S. then it makes it undetermining to go in which direction.

(iii) Ridge: higher than the surrounding areas & itself has a slope & make it impossible to traverse the ridge in a single move



Solution: (i) Backtrack (ii) Make a big jump (in search space).



(iii) Apply two or more rules before doing the test,

① * Example Hill Climbing :

A
H
G
F
E
D
C
B

initial state

H
G
F
E
D
C
B
A

goal state

- Add one point for every block that is residing correctly on which it is supposed to.
- subtract one point for every block that is sitting wrongly.

H
G
F
E
D
C
B

A

Hill climbing will accept this move as it is better. Here we have further 3 possible ~~to~~ moves as below.

A
H
G
F
E
D
C
B

(a)

G
F
E
D
C
B

(b)

H
A

G
F
E
D
C
B

(c)

• Hill climbing will halt here.
⇒ It is a local maximum problem.

* Global :

- for every correct structure add one point and subtract one point for every incorrect structure.

-7	A
-6	H
-5	G
-4	F
-3	E
-2	D
-1	C
0	B

-28 IS

7	H
6	G
5	F
4	E
3	D
2	C
1	B
0	A

-28 GS

H
G
F
E
D
C
B
A

-91

-28

-5	G
-4	F
-3	E
-2	D
-1	C
0	B

(b)

-16

H
A

(c)

-5	G
-4	F
-3	E
-2	D
-1	C
0	B

(a)

-15

Best First search : (Control strategy).

(P48)

- follow a single path at a time but switch path whenever some competing path looks more promising than the current one does.
- It combines the advantages of both DFS & BFS i.e.
 - don't have to explore all branches (DFS).
 - does not get trapped in a dead end (BFS).

